TRAVEL MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

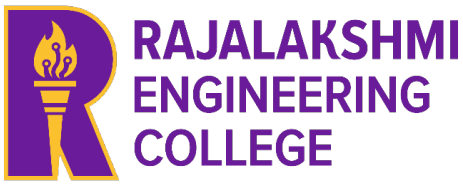
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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TRAVEL MANAGEMENT SYSTEM

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Submitted for the End Semester Examination (Individual Project) held on 10 November 2026.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

TRAVEL MANAGEMENT SYSTEM

I, Manoj Kiraan K (Register No. 240701305), hereby declare that the project report entitled “Travel Management System” is a record of the original work carried out by me on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, under the guidance of the course faculty, in the Department of Computer Science and Engineering, Rajalakshmi Engineering College.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

SIGNATURE OF THE STUDENT

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru. S. Meganathan, B.E., F.I.E., our Chairperson Dr. (Mrs.) Thangam Meganathan and our Vice-Chairman Thiru. Abhay Shankar Meganathan, B.E., M.S. for providing the infrastructure and facilities that made this work possible.

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I sincerely thank Manoranjini, Head of the Department of Computer Science and Engineering, for the support and departmental facilities provided throughout this project.

I owe my sincere thanks to my course faculty, Dr. J.Jinu Sophia, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

MANOJ KIRAAN K

ABSTRACT

Travelling involves many linked decisions: choosing a destination, fixing a budget, reserving hotels, booking buses, flights or trains, and keeping track of what is spent along the way. In the existing process these details are kept in personal notes, spreadsheets, messaging chats and several booking websites. As a result there is no single place to see a complete trip, no running comparison between the estimated budget and the amount already booked, and no organised record of traveller feedback for future planning.

The Travel Management System (TMS) is a role-based application built on the Zoho Creator low-code platform to solve this problem. Its modules are Trip Planners, Hotel Booking, Transportation Masters, Hotel Masters, Destination Masters, Expense Trackers, Feedback and User Profiles, all reached from a left navigation panel. A dashboard summarises Total Budgets, Total Booking Amount, the number of bookings by transport type (buses, flights and others) and the estimated budget by travel category (Solo, Family and Couple). Deluge scripts validate each submission, calculate hotel nights and booking totals, compare totals with the estimated budget, and drive scheduled reminders, status updates and feedback requests. Role-based access gives Admin, Travel Manager and Traveller users only the menus, records and actions they need.

The application was designed with functional and negative test cases covering validation, booking calculation, budget checks, expense tracking and feedback, together with a deployment, security and governance plan. TMS gives travellers and managers one searchable source of trip information, a live view of budget against bookings, and a clear record of experience, while demonstrating how low-code development can deliver a practical workflow application quickly.

Keywords: Travel Management System; Zoho Creator; Deluge; Workflow Automation; Trip Planning; Hotel Booking; Budget Tracking; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	Travel Management System (TMS)
Application Domain	Travel and tourism – trip planning, booking and expense management
Student Name & Register No.	MANOJ KIRAAN K – 240701305
Institution	Rajalakshmi Engineering College, Thandalam, Tamil Nadu
Course Code and Title	CB23F35 – Low-Code Application Development using Zoho Creator
Platform	Zoho Creator – web application with mobile access objective
Primary Users	Admin, Travel Manager, Traveller
Core Modules	Trip Planners, Hotel Booking, Transportation Masters, Feedback, Expense Trackers
Master Data Entities	Hotel Masters, Destination Masters, User Profiles
Visible Dashboard	Summary cards (Total Budgets, Total Booking Amount), Transport Type counts, Travel Category by Estimated Budget chart
Key Automation	Validation, booking total calculation, budget check, expense tracking, reminders, status update, feedback request
Evidence Supplied	1 photograph of the running Zoho Creator dashboard and navigation panel
Academic Year	2026–2027
Report Date	10 November 2026
Implementation Status	Navigation, dashboard summary cards and charts visible; forms, automation and permissions require verification

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
ID	Identifier
JPG	Joint Photographic Experts Group
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PDF	Portable Document Format
PNG	Portable Network Graphics
RBAC	Role-Based Access Control
TMS	Travel Management System
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Individuals, couples and families plan trips by choosing a destination, setting an estimated budget, selecting a hotel and booking transport such as buses, flights and trains. Costs are then incurred during the trip and the experience is rarely recorded in a structured way. The kind of trip also changes the plan: a solo traveller, a couple and a family have different budgets, room needs and transport preferences. TMS provides a central application in which each trip is stored with its destination, travel category, dates, estimated budget, bookings, expenses and feedback. The system is designed to reduce dependence on scattered notes, spreadsheets and chat messages.

1.2 Problem Statement

The existing planning process does not consistently provide one place to see a complete trip, a running comparison of estimated budget against booked amounts, or a structured record of expenses and feedback. As a result, bookings may exceed the budget without being noticed, good hotels and routes are forgotten, and managers or families cannot easily produce a reliable summary of what a trip cost.

- Trip details, hotel confirmations and transport tickets are spread across chats, e-mails and booking websites.
- Without a running total, it is unclear how much of the estimated budget has already been committed.
- Expenses and traveller feedback are not captured in a searchable form that can inform later trips.

1.3 Objectives

- To design structured data forms for trips, hotel bookings, transport, hotels, destinations, expenses, feedback and users.
- To specify and implement validation, booking-total calculation, budget checks, reminders, status updates and feedback requests.
- To provide a dashboard with summary cards and charts, and reports for each module, with role-specific access rules.
- To use Zoho Creator email capabilities to notify travellers and managers about reminders and budget alerts.
- To document secure deployment, validation, audit and governance controls for the application.

1.4 Scope of the Project

In scope: trip planning; destination, hotel and transport masters; hotel booking; transport records; estimated budgets and booking totals; expense tracking; feedback; email notification; dashboard; reports; and role-based access. Out of scope: online payment processing, live fare and room availability from airlines or hotel chains, visa and insurance processing, replacing third-party booking websites, and guaranteeing external email delivery. The platform is Zoho Creator, accessed through a web browser and intended to be mobile-friendly.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable because TMS is primarily a data-centric workflow application. Forms and lookup relationships represent trips, destinations, hotels, bookings and expenses. Reports and pages provide searchable lists and a dashboard with charts. Deluge can implement validation, calculations and scheduled actions. Built-in hosting and mobile access reduce infrastructure work, while permissions and audit facilities support governance. Low-code does not remove the need for careful data modelling, security configuration and testing [2], [3].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing process commonly begins when a traveller notes down trip ideas. Destinations and hotels are searched separately, and bookings are made through websites, phone calls, agents or messaging. Costs are then tracked by hand in a notebook or spreadsheet. There is no consistent record linking the trip to its hotel, transport and expenses, and no automatic comparison with the budget. TMS formalises these steps through a trip record, booking records and an expense record.

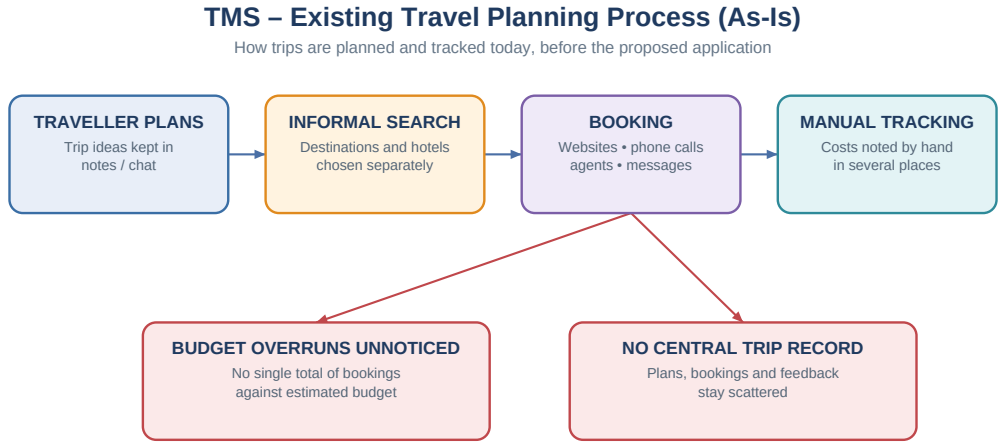


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Information is fragmented across channels and may be duplicated, outdated or overlooked.
- The amount booked is not compared with the estimated budget, so overruns are found late.
- Transport and hotel choices, expenses and feedback require manual collection for any summary or reuse.

2.3 Low-Code versus Traditional Development

Table 2.1: Low-Code versus Traditional Development

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires custom UI, database, authentication and workflow implementation; effort depends on team	Creator forms, reports and workflow builders accelerate the initial application build
Skills required	Frontend, backend, database, hosting and deployment skills	Creator configuration, data modelling and Deluge skills
Hosting & infrastructure	Separate hosting, monitoring, backup and deployment setup	Platform hosting and managed environment, subject to subscription
Workflow changes	Often require code changes and redeployment	Many changes can be made through visual configuration and Deluge
Maintenance	Team maintains application stack and infrastructure	Owner maintains Creator configuration, data, roles and platform subscription
TMS suitability	Suitable for extensive custom integrations, such as live fare search	Suitable for structured trips, bookings, budgets, reports and scheduled automation
Customisation limits	Almost unlimited freedom in interface and integrations	Bound by Creator components, quotas and Deluge capabilities
Cost	Higher staffing and infrastructure effort	Subscription-based; lower build effort; licence to be confirmed

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, current documentation	Zoho Creator Help	Documents forms, reports, pages, workflows and application administration.	Primary platform reference for TMS implementation.
2	Zoho Corporation, current documentation	Deluge Script Reference	Describes Deluge syntax, tasks, collections and workflow scripting.	Supports validation, calculation and notification logic.
3	Zoho Corporation, current documentation	Zoho Creator – Pages and Reports	Explains dashboard pages, report layouts and visualisation features.	Relevant to the dashboard cards and charts.
4	OWASP Foundation	Application Security Verification Standard	Provides verification areas for authentication, access control and data protection.	Informs role and record-level security test design.
5	Pressman and Maxim, 2019	Software Engineering: A Practitioner’s Approach	Explains requirements, design, testing and maintenance practices.	Provides the software engineering basis for the project lifecycle.

2.5 Emerging Trends in Low-Code Development

Relevant low-code trends include visual application assembly, reusable workflow automation, AI-assisted development, integration through APIs and the use of analytics dashboards. For TMS, visual forms and reports support rapid iteration. Workflow automation reduces repetitive reminder and status tasks. Analytics can show how budgets are used across travel categories. AI-assisted development may help draft or explain scripts, but generated logic must be reviewed and tested before use. Personal traveller data should not be sent to unapproved AI services.

2.6 Proposed System

The proposed TMS stores trips centrally and connects each trip to its traveller, destination, hotel booking, transport and expenses. Travellers plan trips and record bookings; the system totals the bookings and compares them with the estimated budget. Travel Managers review over-budget trips. Scheduled workflows remind travellers before departure, update trip status and request feedback after the trip. Admin maintains users, destinations, hotels and transport masters.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Admin	Application administrator with full access.	Manage users, destinations, hotels, transport masters, settings, all trips and reports.
Travel Manager	Reviewer of budgets and bookings.	Monitor trips and bookings; review over-budget trips; view dashboard and reports.
Traveller	Primary user who plans and books trips (solo, couple or family).	Create and edit own trips; record hotel and transport bookings and expenses; give feedback.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall capture trip name, traveller, destination, travel category (Solo/Family/Couple), start date, end date and estimated budget as mandatory fields.	Admin, Traveller	High
FR-02	The system shall validate that the end date is not before the start date, restrict past start dates for non-admin users, require a positive budget and prevent duplicate traveller/destination/start-date trips.	All	High
FR-03	The system shall allow a hotel to be selected from the Hotel Master for the trip destination and shall calculate nights and the booking amount from the nightly rate and rooms.	Traveller, Admin	High
FR-04	The system shall record transport (bus, flight or other type) for a trip with route, date and fare, and count bookings by transport type.	Traveller, Admin	High
FR-05	The system shall total all hotel and transport amounts for a trip and flag the trip for manager review when the total exceeds the estimated budget.	System, Travel Manager	High
FR-06	The system shall record expenses by category against a trip and show spend against budget.	Traveller	Medium
FR-07	The system shall collect a rating and comments for a completed trip, hotel or destination.	Traveller	Medium
FR-08	The system shall show a dashboard with total budgets, total booking amount, transport type counts and budget by travel category, and shall send reminders before departure.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	The dashboard and common reports should load within an institutionally agreed response target under expected record volume; measure during UAT.
NFR-02	Security	Users shall access only the records and actions permitted by their role; a traveller shall not see another traveller's trips.
NFR-03	Usability	The interface shall be understandable to a new user and usable on desktop and supported mobile devices.
NFR-04	Reliability	Scheduled workflows shall be safe to rerun without sending duplicate reminders or recalculating totals incorrectly.
NFR-05	Maintainability	Forms, field link names, workflows and functions shall use consistent naming and documentation.
NFR-06	Accuracy	Booking totals and dashboard figures shall be derived from stored records and recomputed when a booking changes.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (edition/licence to be confirmed); application hosted on the Zoho India data centre (creatorapp.zoho.in)
Scripting language	Deluge
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox, current supported versions
Client devices	Desktop/laptop and supported Android/iOS mobile devices
Connectivity	Internet connection required for the hosted application and email notifications
Data storage	Zoho Creator application data, subject to plan limits
Integration	Authorised email connection; Excel/PDF export features as configured

3.5 Use Case Model

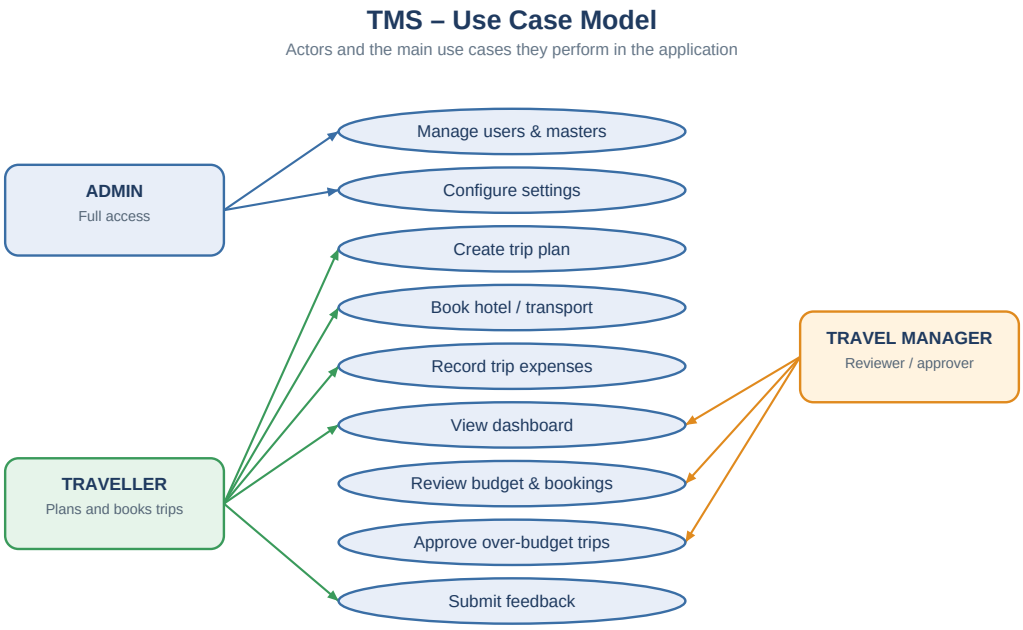


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Plan a Trip and Record Bookings

Use Case ID	UC-01
Use Case Name	Plan a trip and record bookings
Actor(s)	Traveller
Pre-condition	User is signed in and has an active profile; destination, hotel and transport master records exist.
Main Flow	Open Trip Planners; enter trip name, destination, travel category, dates and estimated budget; submit; add hotel booking and transport details; review the totals on the dashboard.
Post-condition	Trip is saved as Planned; booking totals are updated and reflected on the dashboard.
Alternative Flow	If validation fails, submission is blocked and the user receives a field-level message.
Exception	If the booking total exceeds the estimated budget, the trip is flagged and the Travel Manager is notified for review.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The architecture has four logical layers. The presentation layer contains Creator forms, reports, the dashboard page and mobile views for each role. The application layer contains validation rules, Deluge functions and scheduled workflows. The data layer stores Trip, Hotel Booking, Transportation, Expense, Feedback, Hotel, Destination and User Profile records. The integration layer handles email notifications and authorised exports. A user action is checked against role and record scope before it changes data.

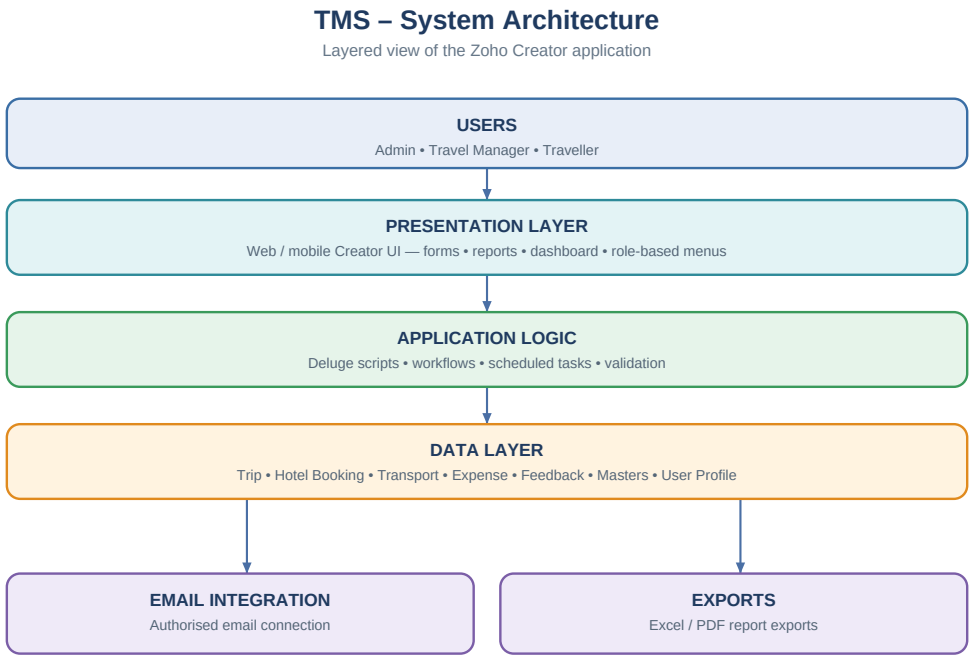


Figure 4.1: System Architecture of the Application

A user signs in through the Creator interface. The application presents only the modules and records allowed for that role. Form submissions invoke validation and workflow logic. Successful actions update the related data records and the trip totals. Email notifications are sent after the relevant event. Dashboard cards and charts read the stored records and aggregate them into indicators. The design should use authenticated record links and avoid exposing traveller details through public URLs.

4.2 Data Model

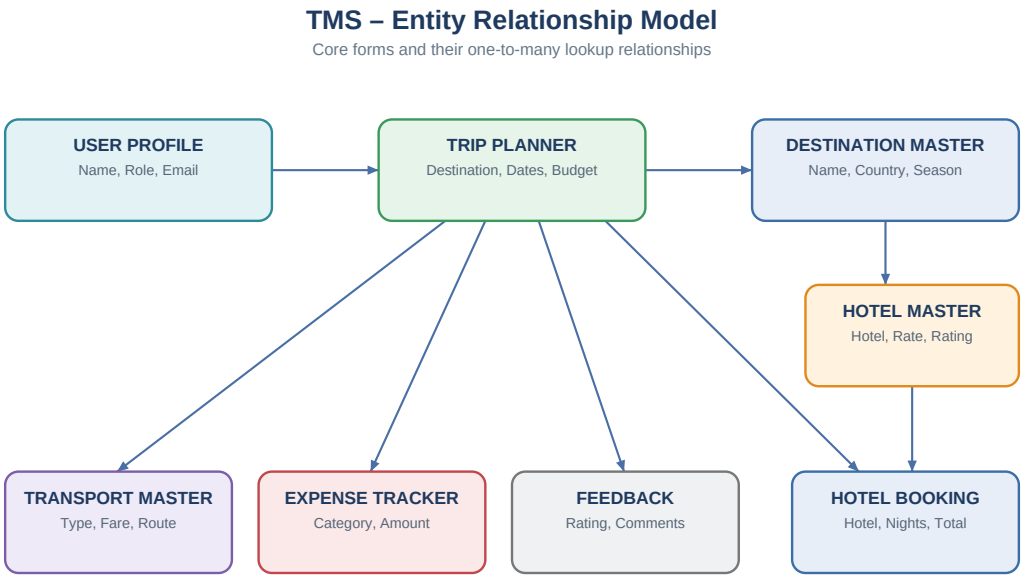


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Trip Planner	Trip ID, name, traveller, destination, category, dates, estimated budget, status	Hotel Booking; Transportation; Expense; Feedback	One-to-Many	Lookup in child forms
Hotel Booking	Trip, hotel, check-in, check-out, rooms, nights, total	Trip Planner; Hotel Master	Many bookings to one trip/hotel	Lookup fields
Transportation Master	Transport type, operator, route, date, fare, trip	Trip Planner	Many rows to one trip	Lookup field
Expense Tracker	Trip, category, amount, date, notes	Trip Planner	Many expenses to one trip	Lookup field
Hotel / Destination Masters	Name, destination, rate, rating, country	Destination Master; Hotel Master	One-to-Many	Lookup fields

4.3 Form Design

The core forms are Trip Planner, Hotel Booking, Transportation Masters, Hotel Masters, Destination Masters, Expense Tracker, Feedback and User Profile, matching the sections of the navigation panel. The Trip Planner form requires a unique ID, trip name, traveller, destination, travel category, dates, estimated budget and status. Hotel and transport records require a trip lookup plus dates and amounts. The Expense Tracker and Feedback forms require a trip lookup, a category or rating and a date. User Profile requires name, email, phone, role and active status.

Table 4.2: Field Specification – Trip Planner Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Trip ID	Trip_ID	Auto Number	System	Unique	Generated by Creator
Trip Name	Trip_Name	Single Line	Yes	Not blank	Visible in reports
Destination	Destination	Lookup	Yes	Must reference Destination Master	Add master record if absent
Travel Category	Travel_Category	Drop Down	Yes	Solo / Family / Couple	Used by dashboard chart
Start / End Date	Start_Date, End_Date	Date	Yes	End not before start	Past start blocked for non-admin
Estimated Budget	Estimated_Budget	Currency	Yes	Greater than zero	Source of Total Budgets card
Trip Status	Trip_Status	Drop Down	Yes	Planned (default)	Planned to Reviewed

4.4 Process and Workflow Design

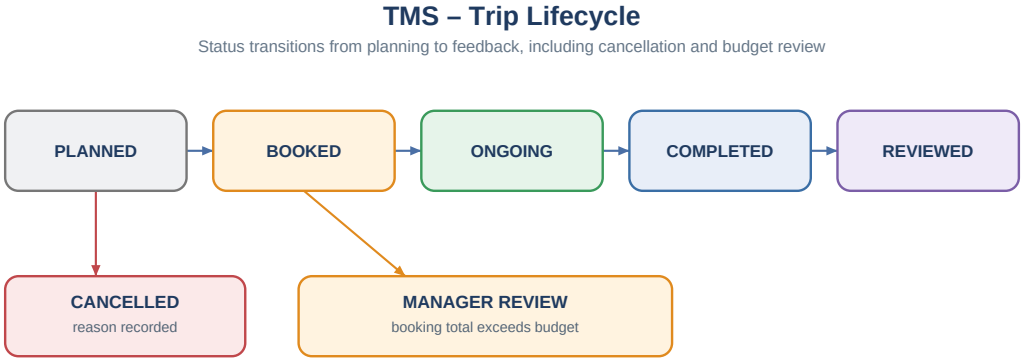


Figure 4.3: Workflow / Process Flow for Trip Lifecycle

4.5 User Interface and Navigation Design

The application uses a dark left navigation panel with a highlighted active item and expandable sections, and a light content area with purple summary cards. The sections visible in the application are Dashboard, Trip Planners, Hotel Booking, Transportation Masters, Feedback, Hotel Masters, Destination Masters, Expense Trackers and User Profiles. The signed-in user’s name appears at the bottom of the panel.

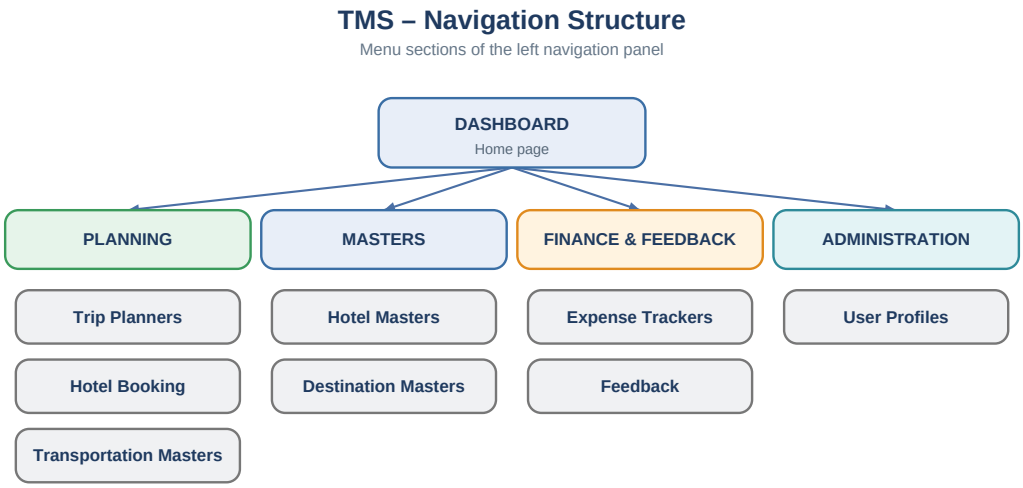


Figure 4.4: Navigation Structure and Wireframe of the Application

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	Admin	Travel Manager	Traveller
Trip Planner	A E V D	V all; review over-budget	A E V own
Hotel Booking	A E V D	V all	A E V own trips
Transportation Masters	A E V D	V all	A E V own trips
Expense Tracker	A E V D	V all	A E V own
Feedback	A E V D	V all	A V own
User Profile	A E V D	V limited	V own profile only
Hotel / Destination Masters	A E V D	V	V

CHAPTER 5
IMPLEMENTATION

The current application is titled “Travel Management System”. The supplied photograph of the running application shows a Creator workspace with a dark left navigation panel. The visible menu contains Dashboard, Trip Planners, Hotel Booking, Transportation Masters, Feedback, Hotel Masters, Destination Masters, Expense Trackers and User Profiles, each with an expand arrow. The application uses a purple and dark-blue colour scheme. Only one photograph was supplied, so implementation details beyond the dashboard and menu are presented as the design to be verified in the live application.

5.1 Application Setup

The application is hosted in Zoho Creator under the address creatorapp.zoho.in, in the section travel-management-system. The left navigation is organised by module, and the Dashboard is the landing page. Any remaining configuration must be verified in the running Zoho Creator application before final submission.

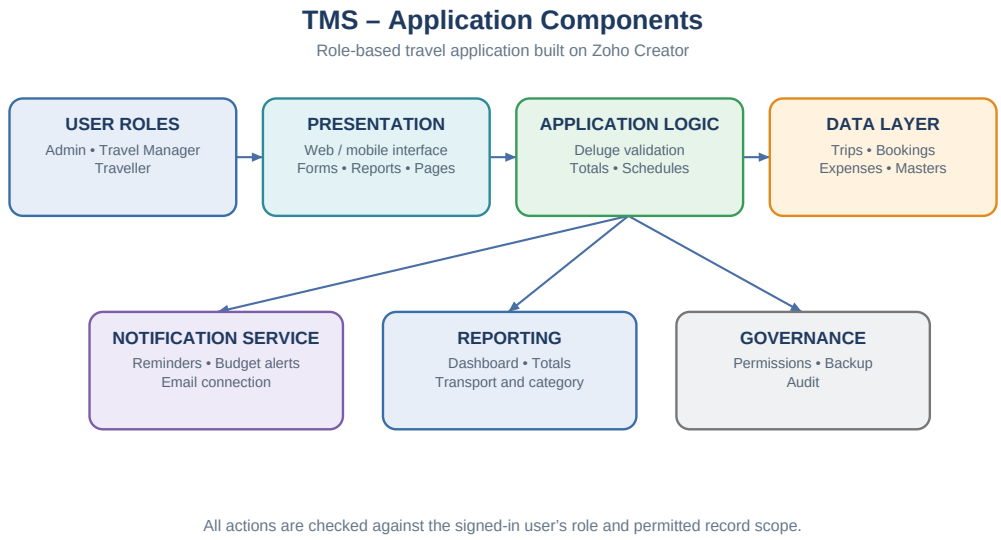


Figure 5.1: Application Components and Navigation Sections

5.2 Forms and Data Capture

Each navigation section opens the forms and reports of one module. The Trip Planner form captures the trip name, destination, travel category, dates and estimated budget. Hotel Booking captures the hotel, check-in and check-out dates, rooms and calculated amount. Transportation Masters records the transport type (bus, flight or other), route, date and fare. Expense Trackers records spending by category. Feedback captures a rating and comments. Hotel Masters and Destination Masters hold the reference data, and User Profiles holds each user's name, email, phone, role and active status. The field specification in Table 4.2 is the design basis; additional fields should be added to the existing forms without creating duplicate forms.

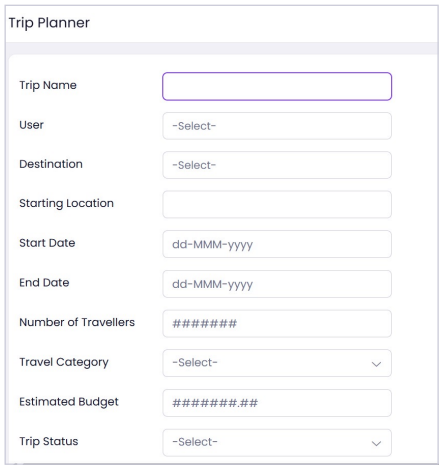


Figure 5.2: Trip Planner Form in Zoho Creator

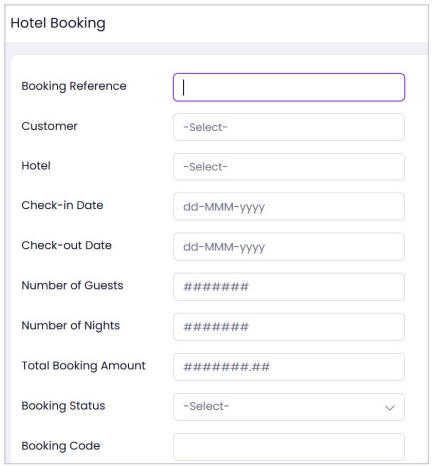


Figure 5.3: Hotel Booking Form in Live Mode

5.3 Reports

Each module has a list report. The final reporting set should include trips by status and category, hotel bookings by destination, transport bookings by type, expenses by category and trip, feedback ratings by hotel, and trips that exceed their estimated budget. Role-specific criteria should be applied to every report so that travellers see only their own trips.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Trips	List	Trip Planner	Filter by status, category and date; sort by start date	Open / Edit within permission	Admin, Manager, Traveller (own)
Hotel Bookings	List	Hotel Booking	Group by destination; sort by check-in	Open linked trip	Admin, Manager, Traveller (own)
Transport Bookings	Grouped	Transportation	Group by transport type	Open record	Admin, Manager, Traveller (own)
Expenses	List	Expense Tracker	Filter by trip and category	Export	Admin, Manager, Traveller (own)
Feedback	List	Feedback	Sort by rating	Open record	Admin, Manager
All User Profiles	List	User Profile	Filter active status and role	Admin maintenance	Admin

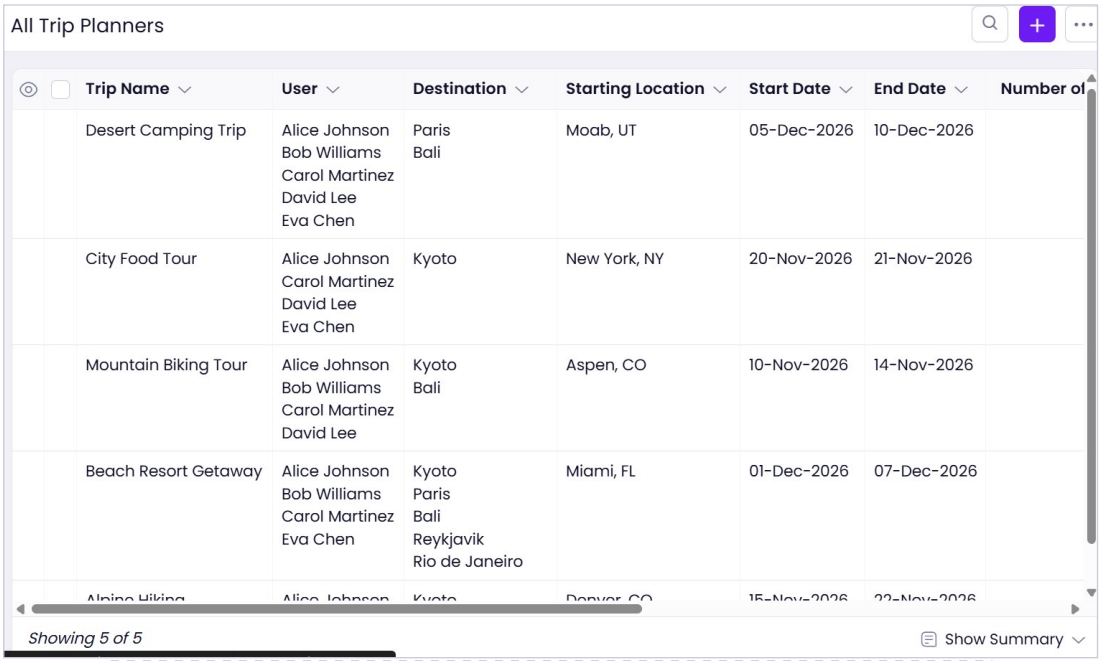


Figure 5.4: All Trips Report

5.4 Pages and Dashboards

The dashboard photograph shows summary cards, a transport section and a chart. The first card, **Total Budgets**, shows $12,800.00. The second, **Total Booking Amount**, shows $7,566.50, and a further card is cut off at the edge of the photograph. Under **Transport Type** the dashboard shows 0 Buses and 2 Flights, with a third transport tile partly visible. Below, the bar chart **Travel Category by Estimated Budget** compares Solo, Family and Couple trips on a scale running from 0 to about 4,000, with the Solo bar extending beyond the visible area.

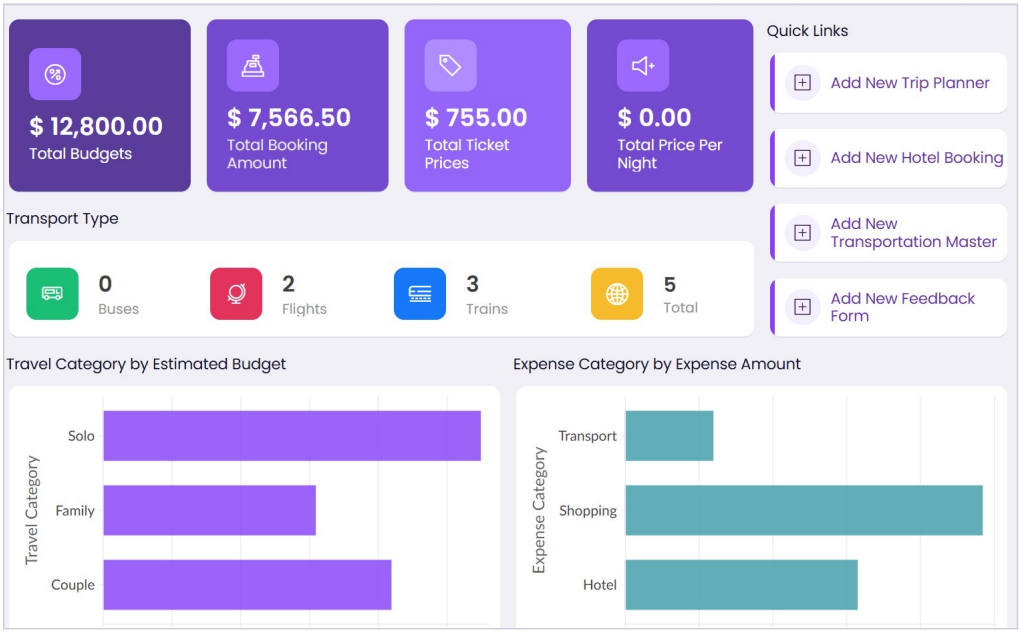


Figure 5.5: TMS Dashboard (photograph of the running application)

The final design should provide separate dashboard views for Traveller, Travel Manager and Admin. Traveller pages should prioritise upcoming trips, remaining budget and pending feedback. Manager pages should show over-budget trips awaiting review. Admin pages should show category, destination and monthly statistics.

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Trip Planner → On Validate	Validate required fields, dates, budget and duplicate trips.	if/else, criteria fetch, alert, cancel submit
DS-02	Hotel Booking → On Validate / On Success	Check dates and hotel destination; calculate nights and amount.	lookup, date arithmetic, update
DS-03	Function: Update_Trip_Totals(Trip_ID)	Total hotel and transport amounts; flag over-budget trips and notify.	criteria fetch, sum, if/else, sendmail
DS-04	Expense Tracker → On Success	Compare spend with budget; warn at 90 percent.	sum, if/else, sendmail
DS-05	Scheduled workflows	Remind before departure, update trip status and request feedback.	schedule, criteria, date arithmetic

Code 5.1: Trip validation script (DS-01, extract)

```
// DS-01 | Form: Trip_Planner | Event: On Validate
if(input.End_Date < input.Start_Date)
{
    alert "End Date cannot be earlier than Start Date.";
    cancel submit;
}
if(input.Estimated_Budget == null || input.Estimated_Budget <= 0)
{
    alert "Estimated Budget must be greater than zero.";
    cancel submit;
}
```

Code 5.2: Budget check script (DS-03, extract)

```
// DS-03 | Function Update_Trip_Totals(int tripId)
total = Hotel_Booking[Trip == tripId].sum(Total_Amount)
      + Transport_Booking[Trip == tripId].sum(Fare);
trip = Trip_Planner[ID == tripId];
trip.Total_Booking_Amount = total;
trip.Over_Budget = (total > trip.Estimated_Budget);
```

The intended Deluge logic includes server-side validation, duplicate detection, booking-total calculation, budget comparison and notification dispatch. The exact Creator field link names and workflow triggers must be confirmed in the live builder before scripts are pasted. No script should be labelled as tested unless it has been executed successfully in the application. The complete scripts are given in Appendix C.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Trip Validation	Form Workflow	On Validate	Required fields and business rules	Block invalid submission	DS-01
Booking Calculation	Form Workflow	On Success	Hotel booking saved	Calculate nights and amount	DS-02
Budget Check	Function	After booking / expense	Total above estimated budget	Flag trip; email traveller and manager	DS-03
Expense Warning	Form Workflow	On Success	Spend at 90 percent of budget	Warn traveller	DS-04
Reminder / Status / Feedback	Schedules	Daily 07:00	Start in 2 days; dates passed	Remind, set Ongoing or Completed, request feedback	DS-05

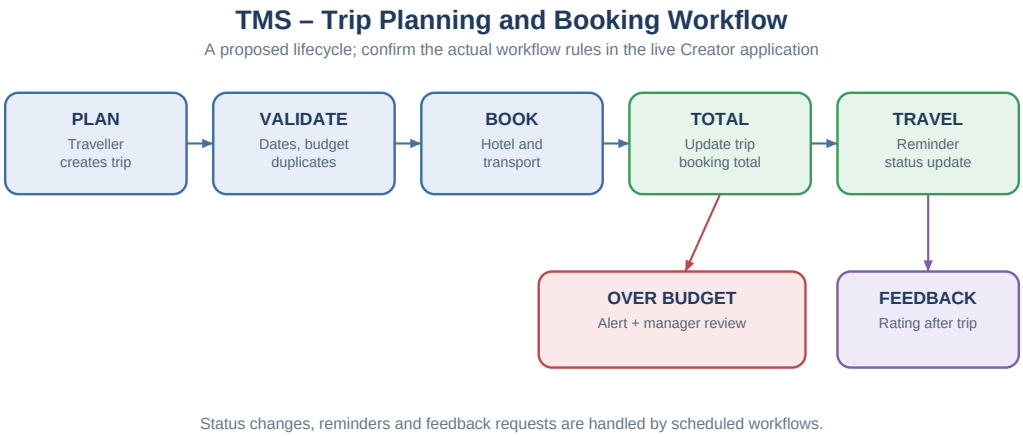


Figure 5.6: Workflow Configuration

5.7 Integrations

The primary integration is email notification through an authorised Zoho Creator connection or supported send-mail action. The application needs messages for budget alerts, departure reminders, expense warnings and feedback requests. Email sender identity, connection permissions and recipient address validity must be checked. The supplied photograph does not demonstrate successful email delivery.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Email notifications	Zoho Creator email / authorised sender	Send mail task or configured connection	Authorised Zoho connection	Budget alerts, reminders, warnings, feedback requests
Excel/PDF export	Creator report export	Built-in report export / configured action	Creator user permission	Trip, booking and expense reports
External travel API (future)	Fare / hotel search API	Creator REST API or webhook	OAuth 2.0 / approved token	Live fares and room availability

Code 5.3: Email integration design

```
// Use an authorised Creator connection or configured send-mail action.
// Never hard-code passwords or API secrets in Deluge source.
// Verify sender identity, recipient address, delivery status and error handling.
// The supplied photograph does not prove that email delivery has been tested.
```

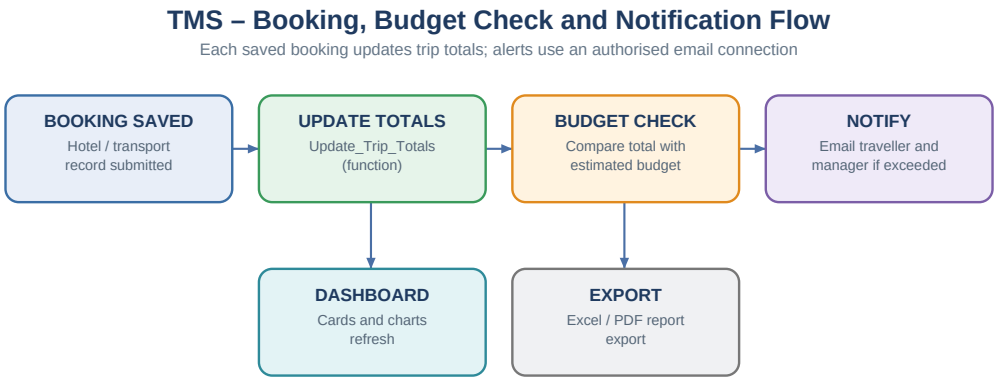
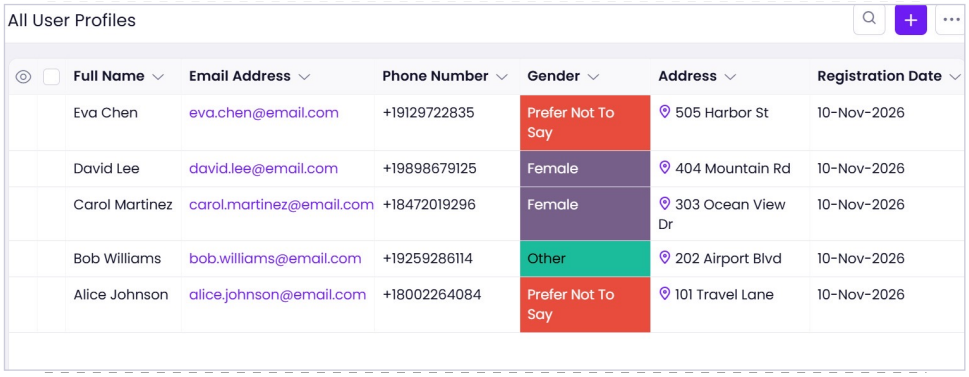


Figure 5.7: Booking, Budget Check and Notification Flow

5.8 User Management and Role-Based Access

Role configuration must enforce Admin full access; Travel Manager read access with review rights for over-budget trips; and Traveller access limited to their own trips and linked bookings, expenses and feedback. Record-level criteria should use the authenticated user and linked profile rather than relying only on hidden menu items. A traveller should not see another traveller's trips, and only Admin should edit master data.

Configure application users with the roles Admin, Travel Manager and Traveller. Map every user to an active User Profile. Use record criteria to restrict travellers to their own trips and child records, and keep the master forms editable only by Admin. The User Profiles section is already present in the navigation panel.



Full Name	Email Address	Phone Number	Gender	Address	Registration Date
Eva Chen	eva.chen@email.com	+19129722835	Prefer Not To Say	505 Harbor St	10-Nov-2026
David Lee	david.lee@email.com	+19898679125	Female	404 Mountain Rd	10-Nov-2026
Carol Martinez	carol.martinez@email.com	+18472019296	Female	303 Ocean View Dr	10-Nov-2026
Bob Williams	bob.williams@email.com	+19259286114	Other	202 Airport Blvd	10-Nov-2026
Alice Johnson	alice.johnson@email.com	+18002264084	Prefer Not To Say	101 Travel Lane	10-Nov-2026

Figure 5.8: User Profiles and Role Configuration Screen

5.9 AI and Analytics Features

The dashboard already displays summary cards, a transport count and a category chart, which provide a basic analytics foundation. The report specification adds budget utilisation, average rating per hotel and counts by destination and month. No AI/Zia feature is evidenced in the supplied photograph. AI-assisted generation may be used during development only if the resulting scripts are reviewed, tested and documented. An AI/Zia feature should not be claimed unless it is visible and demonstrable in the live application.

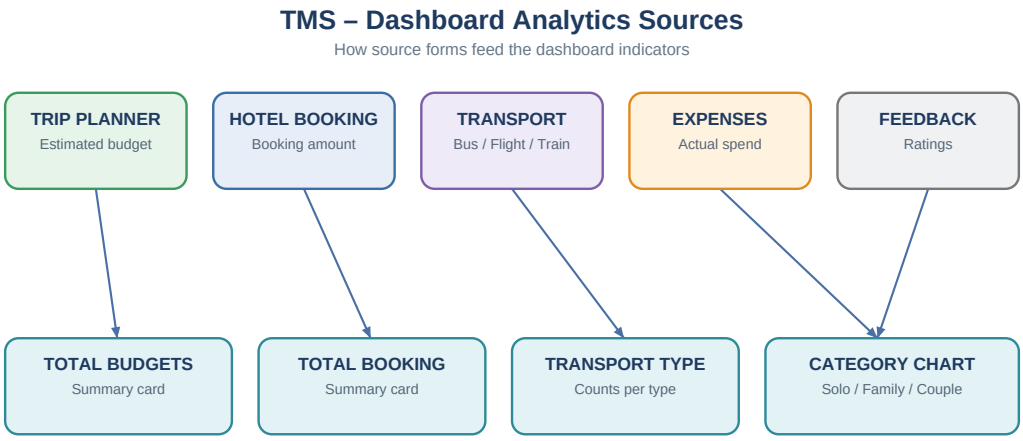


Figure 5.9: Dashboard Analytics Sources

5.10 Pro-Code Extension

No separate pro-code extension is evidenced in the supplied photograph. The current solution is designed around Creator forms, reports, pages and Deluge. If future requirements exceed platform limits, such as live fare search, a Creator REST API, JavaScript widget or Zoho Catalyst service could handle the specialised processing. Such an extension is optional and should be justified by a concrete requirement.

5.11 Mobile and Portal Access

Zoho Creator applications can be opened in a mobile browser or the Creator mobile app. The dashboard photograph shows device-preview icons in the application header, and mobile suitability should be verified on supported devices.

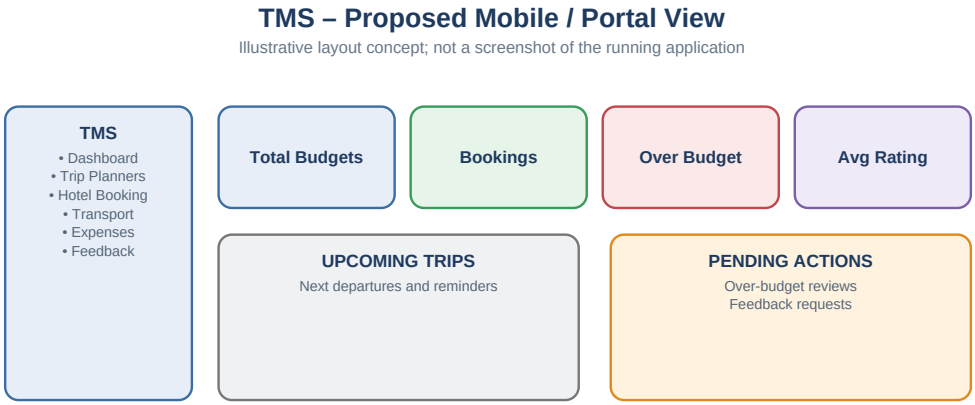


Figure 5.10: Application on Mobile / Customer Portal

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should cover form validation, lookup integrity, booking calculations, budget checks, status transitions, notifications, schedules, exports and access restrictions. The supplied photograph establishes the presence of the dashboard and navigation, but it does not provide execution logs for automation. Therefore, implementation claims should be separated from proposed test expectations. Use separate test accounts for Admin, Travel Manager and Traveller.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit trip with mandatory fields	Enter valid name, destination, category, dates, budget	Trip saved as Planned	To be recorded during live test	Pending
TC-02	FR-01	Submit with trip name missing	Leave name blank	Validation error; no record	To be recorded	Pending
TC-03	FR-02	End date before start date	Set end earlier than start	Submission blocked	To be recorded	Pending
TC-04	FR-02	Past start date as Traveller	Choose yesterday	Submission blocked	To be recorded	Pending
TC-05	FR-02	Zero or negative budget	Enter 0	Submission blocked	To be recorded	Pending
TC-06	FR-02	Duplicate trip	Same traveller, destination, start date	Submission blocked	To be recorded	Pending
TC-07	FR-03	Hotel booking nights and amount	3 nights, rate 100, 2 rooms	Nights 3; amount 600	To be recorded	Pending
TC-08	FR-03	Check-out before check-in	Check-out earlier	Submission blocked	To be recorded	Pending
TC-09	FR-03	Hotel from another destination	Select hotel of other city	Submission blocked	To be recorded	Pending
TC-10	FR-04	Transport type counts	Add one bus and two flights	Dashboard shows 1 bus, 2 flights	To be recorded	Pending
TC-11	FR-05	Total within budget	Bookings below budget	No alert; total updated	To be recorded	Pending
TC-12	FR-05	Total above budget	Bookings exceed budget	Trip flagged; emails sent	To be recorded	Pending
TC-13	FR-06	Expense at 90 percent	Add expenses to 90 percent	Warning sent once	To be recorded	Pending
TC-14	FR-07	Feedback after trip	Submit rating 4	Rating stored against trip	To be recorded	Pending
TC-15	FR-08	Rerun daily schedule	Run twice for same day	No duplicate reminders	To be recorded	Pending

6.3 Debugging Deluge Scripts

Debugging should use Deluge info statements, the editor’s execute/debug facilities, workflow execution history and Creator error messages. For each defect, record the exact form, field link name, event, test record, user role and observed error. A common risk in TMS is an empty lookup value, such as a booking without a trip, causing the total calculation to fail. Another risk is a reminder being sent twice when a daily schedule runs more than once. These cases should be deliberately tested and logged.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Dashboard shows 0 Buses in the Transport Type section.	Dashboard photograph shows 0 Buses and 2 Flights; likely no bus bookings in the sample data.	Add sample bus bookings and confirm the count updates.	To verify
BUG-02	Currency is displayed as dollars on the dashboard cards.	Currency field or format may be set to the default instead of the college's currency.	Confirm the intended currency and standardise the field format.	To verify

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[Tester 1]	Traveller	[Record feedback after live demonstration]	[Record correction or confirm no change]
[Tester 2]	Travel Manager	[Record feedback after live demonstration]	[Record correction or confirm no change]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The deployment process should separate development changes from the live application. Before publishing, confirm form fields, workflows, user permissions, email connections, schedules, report filters and sample-data handling. Use version notes to record changes. Keep a rollback plan and test critical paths after publishing. The institution must confirm the licence and ownership arrangement before production use.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed in Development / Stage	[ ]
2	Sample/test data removed or isolated	[ ]
3	Roles, permissions and users configured in Production	[ ]
4	Email connection and sender identity verified	[ ]
5	All schedules enabled and timezone checked	[ ]
6	Backup and restore procedure confirmed	[ ]
7	Mobile interface tested on supported devices	[ ]

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Filter report records by traveller and status	Trip and booking reports	Unfiltered list may load unnecessary records	Criteria-based view	Lower data volume presented
Recompute totals only for the affected trip	Update_Trip_Totals	Recalculating all trips on each save	Single-trip criteria fetch	Reduced repeated queries
Mark reminders as sent	Daily schedule	Repeated run could resend reminders	Idempotent reminders	No duplicate emails

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role and record-level permissions	All forms and reports	Unauthorised data access
Own-record criteria for travellers	Trips, bookings, expenses	Disclosure of another traveller’s trip and spend
Master data restricted to Admin	Hotel and Destination Masters	Tampering with rates and reference data
Authorised email connections	Notification workflows	Credential leakage and unauthorised sending

7.4 Governance and Administration

Governance includes maintaining accurate user profiles and master data, reviewing role permissions, monitoring failed workflows and bounced email, and defining retention for completed trips. Admin should review inactive accounts and out-of-date hotel rates periodically. Backups and exports must be stored securely. A proposed archival rule moves completed trips older than one year to an archive view, subject to the owner’s retention policy.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A representative scenario begins when a traveller plans a Couple trip with an estimated budget. The form validates the details and saves the trip. The traveller adds a hotel booking and a flight, and the system totals the booked amounts and compares them with the budget. The dashboard shows the updated Total Booking Amount and the flight count. If the total exceeds the budget, the traveller and the manager are alerted. Before departure a reminder is sent, and after the trip a feedback request is issued.

In the supplied dashboard the Total Booking Amount of $7,566.50 compared with Total Budgets of $12,800.00 means about 59.1 percent of the combined estimated budget has been booked, leaving $5,233.50. The Transport Type section shows two flight bookings and no bus bookings. The category chart shows that Family trips have an estimated budget of about 3,100 and Couple trips about 4,200, read from the chart axis, with the Solo bar extending beyond the visible area.

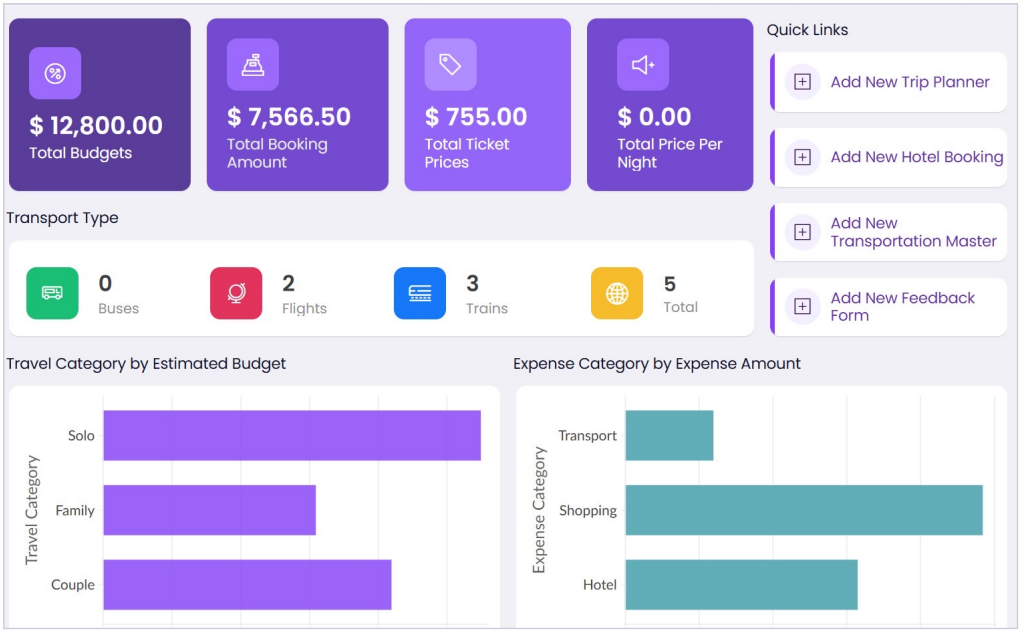


Figure 8.1: Dashboard – Budgets, Bookings and Transport Type

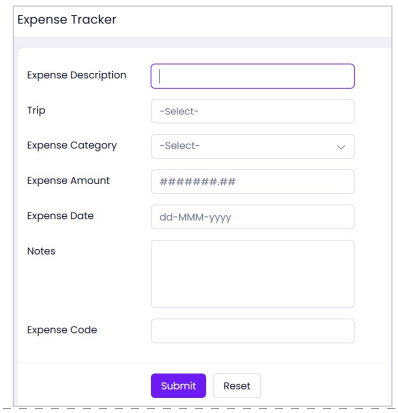


Figure 8.2: Hotel Booking and Expense Records

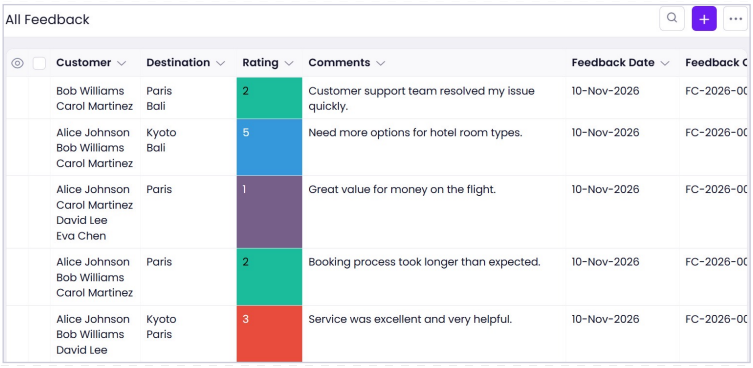


Figure 8.3: Feedback Report

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Structured data model	Trip, Hotel Booking, Transport, Expense, Feedback, Master and User forms	Modules visible in navigation; field details to verify
Automated validation and totals	Deluge workflows and functions	Specified; live execution evidence required
Role-based dashboards and reports	Creator pages and report criteria	Dashboard cards and chart visible; per-role security to verify
Email notification	Authorised email connection	Specified; delivery evidence required

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Trip information	Notes, chats and spreadsheets	Central trip record linked to bookings	Single source of trip data
Budget control	Manual addition of costs	Booking total compared with estimated budget	Early over-budget alert
Expense tracking	Paper or personal sheets	Expense Tracker linked to trip	Measurable spend
Feedback	Informal and forgotten	Feedback form and report	Reusable experience data

8.4 Limitations of the Low-Code Approach

The practical limitations include dependency on the Creator subscription and quotas, the need to maintain Deluge field link names, constraints on highly customised UI behaviour, and the need to configure integrations and role criteria correctly. Live fare and room availability would need an external API. Vendor/platform dependency can be reduced by documenting the data model, workflow logic and export strategy. No specific platform limit or performance benchmark is claimed without measurement.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The visible Creator configuration is largely no-code: form fields, reports, menus and dashboard cards. Deluge represents the low-code layer for validation, calculations, notifications and scheduled actions. A pro-code extension would be justified only for capabilities that cannot be implemented reliably within Creator, such as a live travel-search service or a custom interface component. The project should remain maintainable by the owner’s authorised application administrator.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

TMS establishes a central structure for trips and connects trip details with destinations, hotels, transport, expenses and feedback. The supplied photograph demonstrates the Creator navigation, dashboard summary cards, transport counts and the budget chart. The project specification defines the additional workflows needed for validation, booking totals, budget checks, reminders and feedback. These automations and permissions must be verified through live tests before the application is treated as production-ready.

9.2 Future Enhancements

- Integration with live flight, bus, train and hotel search services through APIs.
- Push notifications, calendar integration and itinerary sharing for families and groups.
- Multi-currency budgets, receipt uploads and automatic expense categorisation.
- Destination recommendations using traveller feedback and Zia analytics.

9.3 Learning Outcomes and Career Relevance

The project develops practical skills in requirements analysis, low-code data modelling, form and report design, Deluge workflow planning, dashboards, access control, testing and deployment governance. These skills are relevant to low-code application development, workflow automation, business analysis and solutions consulting. The work also demonstrates that a visual builder still requires software engineering discipline, especially around validation, security and test evidence.

REFERENCES

[1] W. Rosam and O. Kalnichevski, *Low-Code Development: A Practical Guide*. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 09 November 2026).

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 09 November 2026).

[4] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 09 November 2026).

[5] O. Matvitskyy, K. Davis and A. Jain, “Magic Quadrant for Enterprise Low-Code Application Platforms,” Gartner Research, Oct. 2024.

[6] P. M. Gomes and M. A. Brito, “Low-Code Development Platforms: A Descriptive Study,” in Proc. 17th Iberian Conf. Information Systems and Technologies (CISTI), Madrid, Spain, 2022, IEEE.

[7] Y. Luo, P. Liang, C. Wang, M. Shahin and J. Zhan, “Characteristics and Challenges of Low-Code Development: The Practitioners’ Perspective,” in Proc. ACM/IEEE Int. Symp. Empirical Software Engineering and Measurement (ESEM), Bari, Italy, 2021. doi: 10.1145/3475716.3475782.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Y = visible in the supplied evidence. **Design** = specified in this report and to be verified in the live application. **Future** = planned enhancement, not implemented.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Status	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	TMS navigation menu	5.1	Figs. 5.1, 5.5
II	Forms with multiple field types	Design	Trip, Booking, Expense, Feedback, Master forms	5.2	Figs. 5.2–5.3
II	Lookup field (relationship between forms)	Design	Trip / Hotel / Destination relations	4.2, 5.2	ER model
II	Parent-child relationship	Design	Bookings, expenses, feedback per trip	4.2	ER model
II	Report – List	Design	All Trips / Hotel Bookings	5.3	Fig. 5.4
II	Report – Grouped / category view	Design	Transport Bookings by type	5.3	Table 5.1
II	Report filters and sorting	Design	Report criteria	5.3	Table 5.1
II	Pages and dashboards	Y	Dashboard with summary cards	5.4	Fig. 5.5
II	Chart / data visualisation	Y	Travel Category by Estimated Budget	5.4	Fig. 5.5
II	Deluge form validation	Design	Trip Planner, Hotel Booking	5.5	DS-01, DS-02
II	Deluge custom function	Design	Update_Trip_Totals(Trip_ID)	5.5	DS-03
II	Deluge email notification	Design	Budget alert and reminders	5.5, 5.7	DS-03, DS-05
II	Workflow – On Validate	Design	Trip Planner form	5.6	DS-01
II	Workflow – On Success	Design	Hotel Booking, Expense Tracker	5.6	DS-02, DS-04
II	Scheduled workflow	Design	Reminder, status, feedback request	5.6	DS-05
III	Email integration	Design	Authorised Creator email connection	5.7	Code 5.3
III	External API / webhook integration	Future	Live fare / hotel search	5.7	Table 5.4
III	User management and roles	Y	User Profiles section	5.8	Fig. 5.8
III	Record-level permission criteria	Design	Traveller own-record access	5.8	Permission tests
III	AI / Zia feature	Future	Not evidenced	5.9	—
III	Analytics / dashboard metrics	Y	Summary cards and charts	5.4, 5.9	Figs. 5.5, 5.9
V	Pro-code extension	Future	Not evidenced	5.10	—
V	Mobile app access	Design	Mobile access objective	5.11	Fig. 5.10
IV	Deployment / production publish	Design	Creator deployment settings	7.1	Table 7.1
IV	Security / governance controls	Design	Role and record permissions	7.3–7.4	Table 7.3
IV	Backup / audit and retention	Design	Admin governance	7.4	Fig. 7.2

Summary: 5 of 26 features are evidenced in the supplied screenshot, 18 are specified for verification in the live application, and 3 are future work.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development.	Ch. 1, 2, 4	[ ]	[ ]
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5, 6	[ ]	[ ]
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs and connections.	Ch. 5.7	[ ]	[ ]
CO4	Apply user management, security, deployment and governance controls.	Ch. 4.6, 7	[ ]	[ ]
CO5	Evaluate low-code limitations and identify pro-code extension opportunities.	Ch. 8, 9	[ ]	[ ]

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Field link names are the design names used in this report and must be matched to the live Creator forms before the scripts are used. No script is claimed as tested until it has been executed in the application.

C.1 Trip Planner Form – On Validate (DS-01)

```
// Script ID : DS-01
// Form : Trip_Planner     Event : On Add / On Edit > Validate on submit
// Purpose : Validate mandatory fields, dates, budget and duplicate trips.
if(input.Trip_Name == null || input.Trip_Name.trim() == "")
{
    alert "Trip Name is required.";
    cancel submit;
}
if(input.Destination == null)
{
    alert "Please select a destination.";
    cancel submit;
}
if(input.Start_Date == null || input.End_Date == null)
{
    alert "Start Date and End Date are required.";
    cancel submit;
}
if(input.End_Date < input.Start_Date)
{
    alert "End Date cannot be earlier than Start Date.";
    cancel submit;
}
// Only Admin may create a trip that starts in the past
loginUser = User_Profile[Email == zoho.loginuserid];
if(loginUser.Role != "Admin" && input.Start_Date < zoho.currentdate)
{
    alert "Start Date cannot be earlier than today.";
    cancel submit;
}
if(input.Estimated_Budget == null || input.Estimated_Budget <= 0)
{
    alert "Estimated Budget must be greater than zero.";
    cancel submit;
}
// Block duplicate trips for the same traveller, destination and start date
dupCount = Trip_Planner[Traveller == input.Traveller && Destination == input.Destination
    && Start_Date == input.Start_Date && ID != input.ID].count();
if(dupCount > 0)
{
    alert "A trip for this traveller, destination and start date already exists.";
    cancel submit;
}
```

C.2 Hotel Booking – On Validate and On Success (DS-02)

```
// Script ID : DS-02
// Form : Hotel_Booking    Event : On Validate, then On Success
// Purpose : Check dates and hotel destination; calculate nights and booking amount.
// ---- On Validate ----
if(input.Check_Out <= input.Check_In)
{
    alert "Check-Out must be after Check-In.";
    cancel submit;
}
hotel = Hotel_Master[ID == input.Hotel];
trip  = Trip_Planner[ID == input.Trip];
if(hotel.Destination != trip.Destination)
{
    alert "The selected hotel is not in the destination of this trip.";
    cancel submit;
}
// ---- On Success ----
rec    = Hotel_Booking[ID == input.ID];
hotel  = Hotel_Master[ID == rec.Hotel];
nights = rec.Check_In.daysBetween(rec.Check_Out);
rec.Nights = nights;
rec.Rate_Per_Night = hotel.Rate_Per_Night;
rec.Total_Amount = nights * hotel.Rate_Per_Night * rec.Rooms;
thisapp.Update_Trip_Totals(rec.Trip);
```

C.3 Function – Update_Trip_Totals (DS-03)

```
// Script ID : DS-03
// Function : Update_Trip_Totals(int tripId)
// Purpose : Total hotel and transport amounts, flag over-budget trips and notify.
void Update_Trip_Totals(int tripId)
{
    trip = Trip_Planner[ID == tripId];
    hotelTotal = Hotel_Booking[Trip == tripId].sum(Total_Amount);
    transportTotal = Transport_Booking[Trip == tripId].sum(Fare);
    total = ifnull(hotelTotal, 0) + ifnull(transportTotal, 0);
    trip.Total_Booking_Amount = total;
    if(total > trip.Estimated_Budget)
    {
        if(trip.Over_Budget == false)   // notify once when the trip first goes over budget
        {
            trip.Over_Budget = true;
            manager = User_Profile[Role == "Travel Manager" && Active == true];
            sendmail
            [
                from :zoho.adminuserid
                to :trip.Traveller.Email, manager.Email
                subject :"[TMS] Budget exceeded: " + trip.Trip_Name
                message :"Booked amount " + total + " exceeds the estimated budget "
                    + trip.Estimated_Budget + " for the trip " + trip.Trip_Name + "."
            ];
        }
    }
    else
    {
        trip.Over_Budget = false;
    }
}
```

C.4 Expense Tracker – On Success (DS-04)

```
// Script ID : DS-04
// Form : Expense_Tracker    Event : On Add > On Success
// Purpose : Compare total spend with the estimated budget and warn at 90 percent.
rec   = Expense_Tracker[ID == input.ID];
trip  = Trip_Planner[ID == rec.Trip];
spent = Expense_Tracker[Trip == trip.ID].sum(Amount);
if(spent >= trip.Estimated_Budget * 0.9 && trip.Expense_Warning_Sent == false)
{
    sendmail
    [
        from :zoho.adminuserid
        to :trip.Traveller.Email
        subject :"[TMS] Expenses near budget: " + trip.Trip_Name
        message :"You have spent " + spent + " of the estimated budget "
            + trip.Estimated_Budget + "."
    ];
    trip.Expense_Warning_Sent = true;   // prevents duplicate warnings on later saves
}
```

C.5 Scheduled Workflow – Reminders, Status and Feedback (DS-05)

```
// Script ID : DS-05
// Schedule : Daily at 07:00 (reminders, status updates, feedback requests)
// 1. Remind travellers whose trip starts in two days (once per trip)
soon = Trip_Planner[Trip_Status == "Booked" && Start_Date == zoho.currentdate.addDay(2)
    && Reminder_Sent == false];
for each t in soon
{
    sendmail
    [
        from :zoho.adminuserid
        to :t.Traveller.Email
        subject :"[TMS] Your trip starts soon: " + t.Trip_Name
        message :"Your trip to " + t.Destination.Destination_Name + " starts on " + t.Start_Date + "."
    ];
    t.Reminder_Sent = true;
}
// 2. Move trips to Ongoing when the start date has arrived
starting = Trip_Planner[Trip_Status == "Booked" && Start_Date <= zoho.currentdate];
for each t in starting
{
    t.Trip_Status = "Ongoing";
}
// 3. Mark trips Completed after the end date and request feedback
finished = Trip_Planner[Trip_Status == "Ongoing" && End_Date < zoho.currentdate];
for each t in finished
{
    t.Trip_Status = "Completed";
    sendmail
    [
        from :zoho.adminuserid
        to :t.Traveller.Email
        subject :"[TMS] How was your trip? " + t.Trip_Name
        message :"Please open TMS and rate your trip, hotel and transport."
    ];
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the TMS application URL provided by the administrator. 2. Sign in with the assigned account. 3. Use the left navigation menu to open the Dashboard and the modules available to your role. 4. On mobile, use the supported Zoho Creator app or mobile browser and verify that the cards, charts and forms fit the screen.

D.2 Instructions for Traveller

- 1. Open Trip Planners and add a trip with name, destination, travel category (Solo, Family or Couple), dates and estimated budget. 2. Submit the trip.
- 3. Open Hotel Booking and Transportation Masters to record your hotel, bus, flight or train. 4. Open Expense Trackers to log spending during the trip.
- 5. Watch the Dashboard for Total Budgets, Total Booking Amount and Transport Type counts. 6. After the trip, open Feedback and rate the trip.

D.3 Instructions for Travel Manager and Admin

- 1. Open the Dashboard and the trip reports to find trips flagged as over budget. 2. Review the bookings and contact the traveller if needed.
- Admin: maintain Hotel Masters, Destination Masters and User Profiles; keep rates and active status current. Verify that every user has the correct role.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student NameMANOJ KIRAAN		KRegister	No.240701305
Project Title	Travel Management System	Date	10/11/2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clear TMS problem, objectives, stakeholders, FR/NFR and use cases	10	
2	System Design	Architecture, data model, form specifications, workflows and permission matrix	10	
3	Forms, Reports & Pages	Trip, booking and expense forms, lookups, reports, dashboard and usability	15	
4	Deluge and Automation	Validation, booking totals, budget checks, schedules and notification logic	15	
5	Integration and Security	Email connection, role/record permissions and data protection	10	
6	Testing and Debugging	Test coverage, negative cases, defect log and UAT evidence	10	
7	Deployment and Governance	Deployment checklist, performance, backup and audit	10	
8	Documentation and Demonstration	Report quality, screenshots, live workflow demonstration	10	
9	Viva-Voce	Conceptual understanding and explanation of design decisions	10	
		TOTAL	100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and supported by clear test evidence; student explains design decisions.
Good	70–89	Feature implemented correctly with minor gaps; explanation and evidence are adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; testing evidence is limited.
Needs Improvement	Below 50	Feature is missing, unreliable or not explained; corrective work is required.

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student NameMANOJ KIRAAN		KRegister No.240701305	
Department / Section	CSE	Date of Viva	10/11/2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1	Explain the TMS trip lifecycle and status transitions.	II		
2	How does the system calculate the Total Booking Amount?	II		
3	How is a trip flagged when it exceeds the estimated budget?	II		
4	How do the master forms and lookup fields connect the data?	II		
5	How would you prevent duplicate reminders from the schedule?	II		
6	How are role and own-record permissions enforced?	III		
7	Which schedules are required and what do they do?	II		
8	What is the role of Deluge in TMS?	II		

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER
Name: _____

EXTERNAL EXAMINER
Name: _____